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Braille Text to Speech Conversion

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***Abstract*— It is well known that communication plays a key role among people. Normal people can easily share their thoughts by reading and writing, but people who are blind, use a different type of script known as Braille which is a model where the blind people touch and feel the raised dots on the braille script. In this way, each language can be characterized using braille script; it just changes from language to language. The aim of this paper is to convert the braille script to text format and later into the speech format so it is easy to read and write the language without learning to use Braille separately. This is also helpful for the common people as they can read and hear letters, books, etc. written by blind people.**

Keywords—python, computer vision, image processing

I. INTRODUCTION

One should communicate with others to know their thoughts to understand and to avoid misunderstandings; that is how communication is important in human lives. People who are blind or visually impaired people might face difficulty in communicating. Louis Braille, the inventor of the braille language, created the language in such way that one can sense it with their fingertips. This helps the visually impaired people to communicate with the common people in a possible way. The project goal is to design and implement a system that will convert the Braille language scripts into text and then into speech, using effective algorithms and techniques.

Few of the objectives that can be achieved through this system are, making the teachers at schools to instruct the blind students and parents can guide their children in their studies or other activities without knowing the braille language. This is how the sighted and blind people reduce the gap between them. As mentioned above, the importance of the project comes from the small and important things. Children shouldn't be separated based on their lack of vision; even the blind children can mingle and read with the normal students with the help of the teacher instructions who might not know the braille language. This system helps to overcome this problem as the system converts the language or the script to text as well as to the speech format. Also, the parents who have no knowledge on this language wish to know their children's study progress and can overcome this easily using this proposed model. In this way, the model is designed making sighted and blind people to have a grip on the language. Moreover, this system can convert the braille language not only to one particular language rather to many useful languages and hence useful for many regional people.

Paper [2] proposed a system where braille was employed inside the body with the help of the micro vibration motor. Six micro vibration motors were used for distinct body parts where the input was taken from the SMS text that was converted into the braille. However, the chances of rate of the errors are high in the system. Optical recognition of Braille was proposed in [3]. The goal of it was to develop a system that converts among some fixed limitations. The input braille image was scanned and the output was in the text format on the user's computer. As technology improves, the system model can be also used in mobiles instead of PCs.

II. BACKGROUND

Braille is a language used by blind people to read and write where they feel the raised dots on the braille sheet with the help of their fingertips. The inventor of the language is Louis Braille (shown in fig. 2.1), and the name of the language is

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named after him. He created this in the nineteenth century and the idea was obtained from the military code language used to send the secret messages. After this, the braille code language has become widely used among many blind people due to its usability and simplicity in reading and writing. The Braille language was translated into other languages like English and Arabic. The Braille script has 6 raised dots that placed in three rows and two columns where 1 to 3 are arranged from top to bottom on the left, 4 to 6 from top to bottom on the right as depicted in fig. 2.

Fig. 2.1 Louis braille.

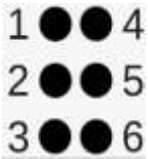


Fig. 2.2 Braille dots.

These six dots are to be raised according to the characters (63 different characters) within a cell by using these dots. This also avoids one to one correspondence between braille cell and word. The dimensions of the braille have been set according to the person's fingertips. The distances between dots in a cell and between cells vary horizontally and vertically. Each dot has approximately 0.5 mm in height, the blank space between dots on adjacent cells is approximately 3.75 mm horizontally and 5.0 mm vertically, the horizontal and vertical spacing between dot centers within a Braille cell is approximately 2.5 mm. A regular Braille page is 11 inches by 11.5 inches (292.1 mm) and usually features a most of 40-43 Braille cells per line in 25 lines. From the 6 dots, the essential braille alphabets, numbers, punctuations, and a few special characters, are square measure build and square measure positioned just like the structure six on a die, in a grid of two parallel vertical lines of each 3 dots. 64 different configurations can be created from a basic six dots grid. These 64 braille characters are seen in uniblock block for 6 dots i.e., U+2800 – U+28FF.

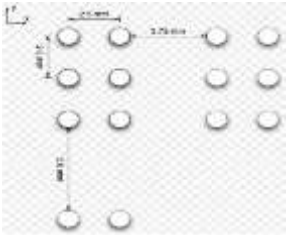
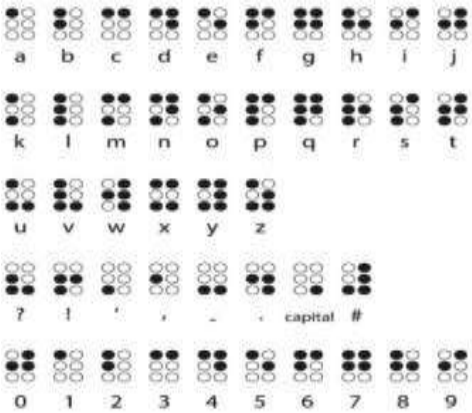


Fig. 2.3 Braille dots dimensions.

Fig. 2.4 Braille alphabets.



III. METHODOLOGY

The Braille text to speech conversion is a software application that helps in converting braille letters to text and then to speech, making it easy to understand by common people. This makes communication much easier. Fig.3.1 shows the architecture diagram of the project which depicts the entire project flow. The user uploads a scanned image of a braille document where the image is separated in dots. Recognizing the letters of braille is done using some python libraries. The braille letters are converted into the corresponding english alphabets. The letters are printed in the form of text and then the text is converted into speech using GTTS (Google Text-To-Speech).

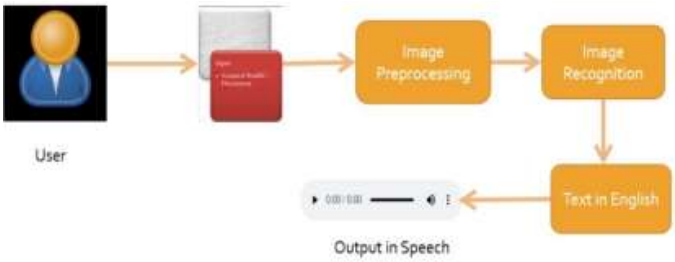


Fig. 3.1 System Architecture Diagram

IV. IMPLEMENTATION

In this implementation phase of Project Management, the activity of the project takes place. Implementation plays a major role including the execution of a project. Some of the software used in building the project are:

- Python that has many libraries and tools for natural language processing. The tools used in this project are imutils and imagieo for image processing.
- Using Python, the detected text is converted into audio/speech with the help of GTTS (Google Text-to-Speech) library.

A. Imutils

Imutils has some functions to make basic image processing, functions such as rotation, translation, resizing, displaying

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Matplotlib images, sorting contours, detecting edges, which makes it easier when compared to OpenCV.

B. Imageio

Imageio is one of the python libraries which is also a cross-platform. Imageio is useful to read and write data of the images in a wide range.

C. Collections

Collection in python is a container with data types, namely sets, tuples, lists, and dictionaries. It has some different characteristics based on the declaration and the usage. Llist is mutable, and it can store duplicate values and elements and can be accessed using indexes. It is declared in square brackets.

D. Opencv

OpenCV is a library for image processing, computer vision and machine learning, which plays an important role in real-time operations. It is mainly used for image processing and to identify objects in videos, handwriting, or even faces of a human.

E. Skimage

Skimage is an open-source Python package used for image preprocessing. It is used to read the given images in a code.

F. Matplotlib Pyplot

Matplotlib.pyplot is a group of functions that make matplotlib work like MATLAB. The pyplot function makes changes to a figure like plotting area with some lines in a figure and some labels are added.

G. Numpy

NumPy -Numerical Python, is a library used for processing those arrays and also for the multidimensional array objects. Using this library, mathematical and logical operations can be performed on arrays.

H. Re

A Regular Expression (or RE) is a set of strings that matches a particular string. The functions in this module lets to check if a given regular expression matches a particular string, which comes down to the same thing.

I. Warnings

Warnings in python are used to warn the developer of situations that aren't necessarily exceptions. Generally, a warning occurs when there are some obsolete programming elements, such as keyword, class or function, etc. This module is actually a subclass of Exception, which is a built-in class in Python.

J. Otsu's Threshold

Otsu's threshold method automatically determines the threshold value but not chosen.

Otsu's Binarization

Consider a photo with handiest wonderful photo values (bimodal photo), wherein the histogram has handiest encompass peaks. A true threshold could be withinside the center of these values. Similarly, Otsu's technique determines an advantageous worldwide threshold fee from the photo histogram.

Algorithm

Step1: Import openCV library.

step2: Read input image using imread.

Step3: Convert image to black and white and blur to remove some of the noise, get edges, and get contours using cv2.

Step4: Sort the contours according to their sizes and apply Otsu's thresholding method to binarize the image.

Step5: Finding the approximate diameter of the contours using cv2.

5.1 for c in ctrs:

5.1.1. peri = cv2.arcLength(c, True)

5.1.2. approx = cv2.approxPolyDP(c, 0.02 * peri, True)

Step6: Sort them based on their positions by calculating area using sorted keyword and by giving key as cv2.contourArea.

6.1 ctrs = sorted (ctrs, key=cv2.contourArea, reverse=True)

Step7: In order to label the contour as a question:

7.1 if diam*0.8 <= w <= diam*1.2and0.8 <= ar <= 1.2:

a. Add to array.

Step8: Sort contours to get the exact positions of dots by choosing the tolerance for x, y coordinates of the bounding boxes to be binned together. Then, change x and y coordinates of bounding boxes to their corresponding bins.

Step9: Get the spaces between two adjacent dots in a letter.

9.1 spacingX = spacing(0)

9.2 spacingY = spacing(1)

9.3 m = min(spacingX)

Step10: A list of alphabets are taken according to the position and numbering of the original braille letters.

Step11: Based on the positions of the above spaces and contours, the dots are given corresponding numbers by checking the numbers and alphabets in the list created.

Step12: Now by considering the above numbers the dots are converted into corresponding English alphabets.

Step13: Display the image using matplotlib and text using a loop iterating through the list.

Step14: The text is converted to speech using Google-Text-to-speech (GTTS) library. An Audio file is displayed using Audio library as shown in fig.5.2.

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V. RESULTS

The following are the results obtained. The fig. 5.1 shows that the braille script image is converted to English text. Here, “thankyou” image of the braille script is considered and the corresponding English text is displayed.

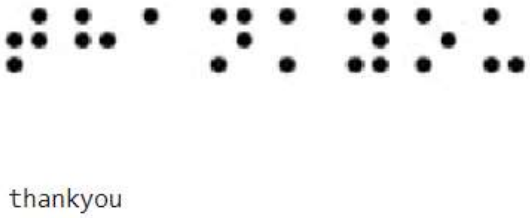

Fig. 5.1 Braille text - English text

The fig. 5.2 shows the audio file that is converted from the text to audio using GTTS. Here, the text input is taken from the output obtained from the above braille script.

Fig. 5.2 Text converted to Audio (using GTTS)

VI. CONCLUSION

The present paper is a flexible and real-time Braille to text and audio conversion system. In This form, the framework can be utilized in a paper, book, archive and so forth. For reading and easy understanding for the impaired audience, another usefulness of Text to Audio has been added to help in understanding books or tuning into book recordings/content. In any case, there are a few upgrades which will be consolidated in future adaptations of the equipment interpreter. It is hoped that this future hi-tech improvement will enlighten the life of the blind population. This software allows social, communicational, and educational barriers to be removed for people with disabilities. Thus, with some modifications in previous conventional communicating devices, a large number of visually impaired people can be accommodated in an updated communication system.

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